$$n = \frac{\sin\left(\frac{60^0 + 37^0}{2}\right)}{\sin\left(\frac{60^0}{2}\right)} = \frac{\sin\left(48.5^0\right)}{\sin\left(30^0\right)} = \frac{0.75}{0.5} = 1.5$$

The refractive index of the material of the prism is, $n = 1.5$

6.7.4 Dispersion of white light through prism

The angle of deviation produced by a prism is so far discussed only for monochromatic light (i.e. light of single colour). When white light enter in to a prism, an effect called *dispersion* takes place. **Dispersion is splitting of white light into its constituent colours. This band of colours of light is called its *spectrum*.** When a narrow beam of parallel rays of white light is incident on a prism and the refracted beam is received on a white screen. A band of colours is obtained in the order, recollected by the word: VIBGYOR i.e., Violet, Indigo, Blue, Green, Yellow, Orange and Red. The violet is the most deviated colour and red is the least deviated colour as shown in Figure 6.43.

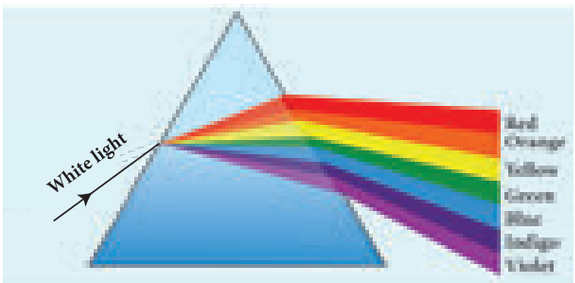


Figure 6.43 Dispersion of white light in to its constituent colours

The colours obtained in a spectrum depend on the nature of the light source. Each colour in vacuum has a definite wavelength. The red colour has the longest wavelength of 700 nm while the violet colour has the shortest wavelength of 400 nm in vacuum. Though all the colours have different wavelengths, they all travel with the same speed in vacuum. The speed of light is independent of wavelength in vacuum. Therefore, vacuum is a non-dispersive medium.

Points to Ponder

Sir Isaac Newton has demonstrated through a classic experiment to produce white light by recombining all the colours of VIBGYOR. He used a prism to produce dispersion and made all the colours to incident on another inverted prism to combine all the colours to get white light as shown in figure.



But, when the white light enters a medium the red colour travels with the highest speed and violet colour travels with least speed. Hence, the wavelengths of colours in a medium are no longer the same as they are in vacuum. Actually, the dispersion takes place in a medium because of the difference in speed for different colours in a medium. In other words, the refractive index of the material of the prism is different for different colours. For violet colour, the refractive index is the highest and for red colour the refractive index is the least. The refractive index of two different glasses for different colours is shown in Table 6.4.

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